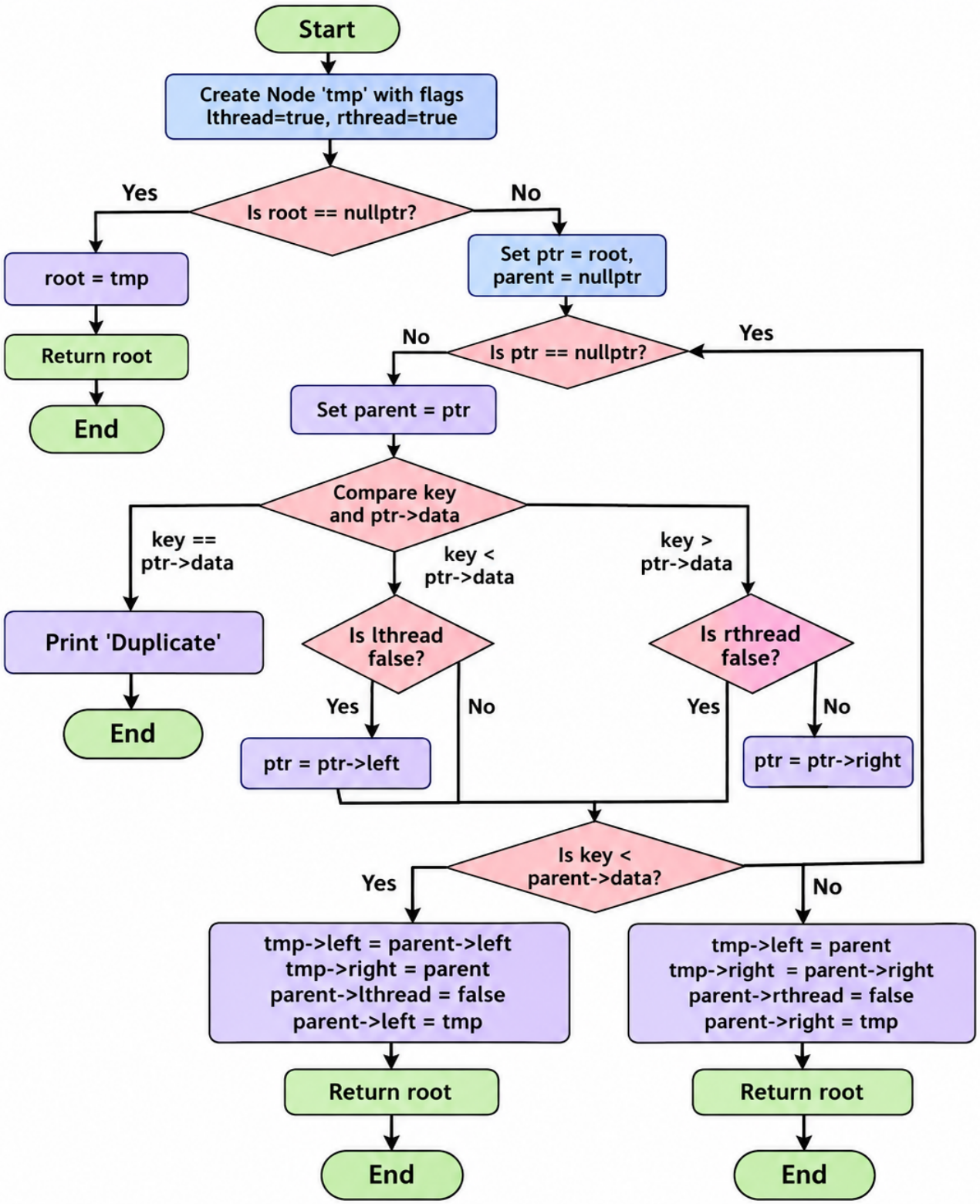

Start
Create Node 'tmp' with flags lthread=true, rthread=true
Is root == nullptr?
Yes
No
root = tmp
Return root
End
Set ptr = root, parent = nullptr
Is ptr == nullptr?
No
Yes
Set parent = ptr
Compare key and ptr->data
key == ptr->data
key < ptr->data
key > ptr->data
Print 'Duplicate'
End
Is lthread false?
Is rthread false?
Yes
No
Yes
No
ptr = ptr->left
ptr = ptr->right
Is key < parent->data?
Yes
No
tmp->left = parent->left
tmp->right = parent
parent->lthread = false
parent->left = tmp
tmp->left = parent
tmp->right = parent->right
parent->rthread = false
parent->right = tmp
Return root
Return root
End
End